Department of Computer Science
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Online Resit Exam for the course *Logic and Sets*
July 9, 2020, 8:30–12:00

This exam has 5 pages and 8 exercises.

The result will be computed as (the total number of points plus 10) divided by 10.

Please motivate all answers!

By participating in this exam, I declare to understand that taking an online exam during this corona crisis is an emergency measure to prevent study delays as much as possible. I know that fraud control will be tightened and realize that a special appeal is being made to trust my integrity. With this statement, I promise to:

- make this exam completely on my own, and not share my solutions with other students,
- not consult sources other than explicitly mentioned in the examination, and
- make myself available for any oral explanation of my answers.

1. Distributivity laws (6 + 6 points)
For each of the two distributivity laws below, regarding conjunction and exclusive or, either argue that it holds or give a counterexample to show that it does not hold.

(a) $(x \wedge y) \oplus z \;\equiv\; (x \oplus z) \wedge (y \oplus z)$

Solution: This distributivity law does not hold. There are two counterexamples, in which z gets the value $\mathtt{T}$, while x and y get opposite truth values (so either x is $\mathtt{T}$ and y is $\mathtt{F}$, or x is $\mathtt{F}$ and y is $\mathtt{T}$).

Then the left-hand side evaluates to $\mathtt{T}$. Namely, $x \wedge y$ is $\mathtt{F}$, because x or y is $\mathtt{F}$. Since moreover z is $\mathtt{T}$, $(x \wedge y) \oplus z$ is $\mathtt{T}$.

On the other hand, the right-hand side evaluates to $\mathtt{F}$. Namely, since x and y have opposite truth values, it follows that $x \oplus z$ and $y \oplus z$ have opposite truth values, so one of them is $\mathtt{F}$. Hence $(x \oplus z) \wedge (y \oplus z)$ is $\mathtt{F}$.

(b) $(x \oplus y) \wedge z \;\equiv\; (x \wedge z) \oplus (y \wedge z)$

Solution: This distributivity law holds. If z is $\mathtt{T}$, then clearly both the left- and the right-hand side or semantically equivalent to $x \oplus y$. And if z is $\mathtt{F}$, then the left-hand side is clearly $\mathtt{F}$, and also the right-hand side is semantically equivalent to $\mathtt{F} \oplus \mathtt{F}$, which evaluates to $\mathtt{F}$.

2. **Logic circuit** (10 points)
Give a logic circuit, containing only AND, OR and NOT gates, that captures the formula $(y \to x) \to \neg(x \to z)$.

Solution: $y \to x$ is semantically equivalent to $\neg y \vee x$.

$x \to z$ is semantically equivalent to $\neg x \vee z$, so $\neg(x \to z)$ is equivalent to $x \wedge \neg z$.

So $(y \to x) \to \neg(x \to z)$ is semantically equivalent to $\neg(\neg y \vee x) \vee (x \wedge \neg z)$, i.e., $(y \wedge \neg x) \vee (x \wedge \neg z)$.

This is captured by the following logical circuit:

```
y  --------> AND -------
              ^        |
              |        |
x  ---> NOT ---        V
    |                  OR --->
    ---------------    ^
                  |    |
                  V    |
z  ---> NOT ---> AND ---
```

3. **DNF + axioms for $\equiv$** (4 + 7 points)

   (a) Provide the truth table for the formula $p \vee q$, and extract a DNF from this truth table that is semantically equivalent to $p \vee q$.

   Solution:

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

   The three lines with a `T` in the third column give rise to the following three disjuncts: $p \wedge q$, $p \wedge \neg q$, and $\neg p \wedge q$.

   So the resulting DNF is $(p \wedge q) \vee (p \wedge \neg q) \vee (\neg p \wedge q)$.

   (b) Derive with the axioms for semantic equivalence that $p \vee q$ and the DNF you extracted in (a) are semantically equivalent.

Solution:

$$
\begin{array}{cll}
 & ((p \wedge q) \vee (p \wedge \neg q)) \vee (\neg p \wedge q) & \\
\equiv & (p \wedge (q \vee \neg q)) \vee (\neg p \wedge q) & \text{(distributivity)} \\
\equiv & (p \wedge \top) \vee (\neg p \wedge q) & \text{(complement)} \\
\equiv & p \vee (\neg p \wedge q) & \text{(identity)} \\
\equiv & (p \vee \neg p) \wedge (p \vee q) & \text{(distributivity)} \\
\equiv & \top \wedge (p \vee q) & \text{(complement)} \\
\equiv & p \vee q & \text{(identity)}
\end{array}
$$

4. Predicate logic (12 points)

Give two models to show that the following two formulas do not semantically entail each other.

- $\exists x\, C(x)$
- $\forall x\, C(x)$

Solution: To prove that the first formula does not semantically entail the second formula, consider a model with elements a, b, where $C(a)$ holds while $C(b)$ does not. The first formula holds on this model, because $C(a)$, but the second formula does not, because $C(b)$ does not hold.

To prove that the second formula does not semantically entail the first formula, consider a model with no elements. The second formula holds on this model, because C holds for all elements (as there are none). But the first formula does not, because there are no elements, so C does not hold for any element.

5. Relations (6 + 4 points)

(a) Is the empty binary relation R_1 in $\mathbb{N}$, which relates no pairs of elements, reflexive? Is it transitive? Is it symmetric? Is it anti-symmetric?

Is the identity binary relation R_2 in $\mathbb{N}$, which relates each element only to itself, reflexive? Is it transitive? Is it symmetric? Is it anti-symmetric?

Solution: R_1 is not reflexive, because $k\, R_1\, k$ does not hold for any $k \in \mathbb{N}$. It is transitive, symmetric, and anti-symmetric, because in each of these cases the "if clause" is vacuous for the empty relation R_1.

R_2 is reflexive, because $k\, R_2\, k$ holds for all $k \in \mathbb{N}$. It is transitive, because $k\, R_2\, \ell$ and $\ell\, R_2\, m$ imply $k = \ell = m$, so that then $k\, R_2\, m$. It is symmetric, because $k\, R_2\, \ell$ implies $k = \ell$, so that then $\ell\, R_2\, k$. It is anti-symmetric, because the "if clause" $k\, R_2\, \ell \wedge k \neq \ell$ is vacuous for the identity relation R_2.

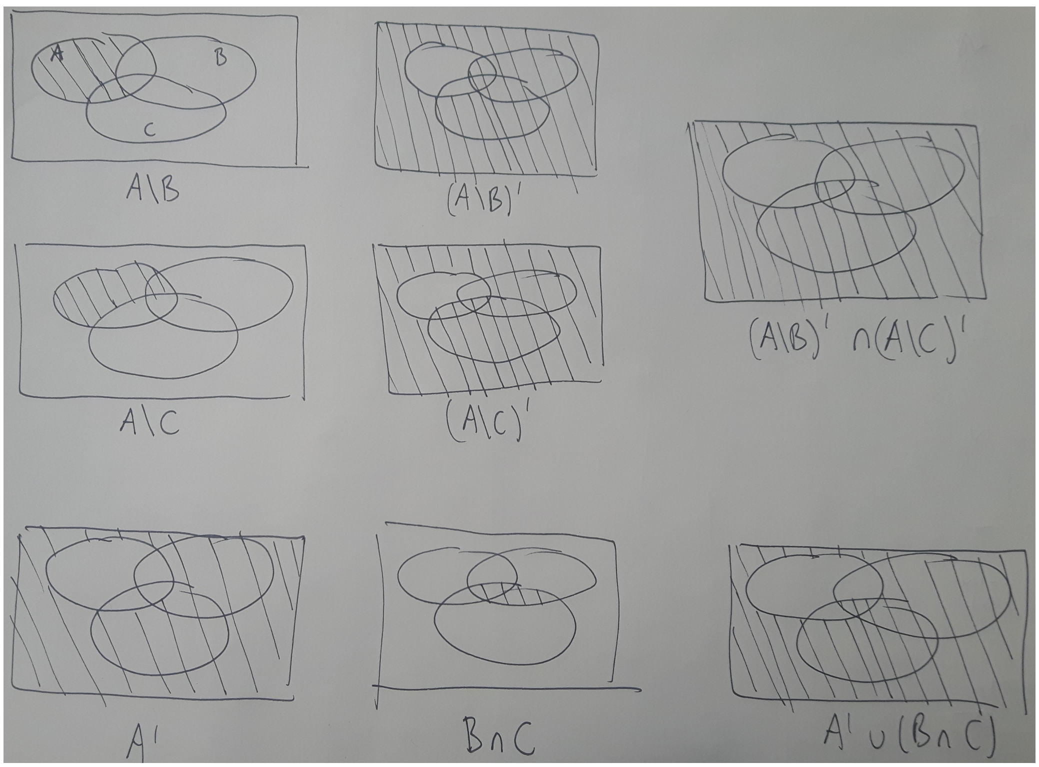
(b) Describe the binary relations $R_1 \circ S$ and $R_2 \circ S$, for any binary relation S in $\mathbb{N}$.

Solution: $R_1 \circ S$ coincides with R_1.
$R_2 \circ S$ coincides with S.

6. Sets (5 + 7.5 points)
Consider the following set-theoretic equality:

$$(A \backslash B)' \cap (A \backslash C)' \;=\; A' \cup (B \cap C)$$

(a) Draw Venn diagrams for the two sets $(A \backslash B)' \cap (A \backslash C)'$ and $A' \cup (B \cap C)$, depicting clearly which area corresponds to the set, and how the set is constructed from A, B and C using the fundamental set operations. Use your Venn diagrams to conclude that the set-theoretic equality displayed above holds.
Solution:



(b) Prove the set-theoretic equality displayed above (for all sets A, B and C) using the rules of the algebra of sets.
Solution:

$$\begin{aligned}
(A \backslash B)' \cap (A \backslash C)' &= (A \cap B')' \cap (A \cap C')' && \text{definition} \\
&= (A' \cup B'') \cap (A' \cup C'') && \text{De Morgan} \\
&= (A' \cup B) \cap (A' \cup C) && \text{involution} \\
&= A' \cup (B \cap C) && \text{distributivity}
\end{aligned}$$

7. **Functions** (6 + 4 points)
We are given the functions $f\colon \mathbb{R} \to \mathbb{R}$, $mul_3\colon \mathbb{R} \to \mathbb{R}$, and $g\colon \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) := \frac{1}{x^2-1} \qquad mul_3(x) := 3x \qquad g(x) := \sqrt{x-1}$$

(a) What is the domain of definition of f, and what is its range?
What is the domain of definition of g, and what is its range?
Solution: The domain of definition of f is $\mathbb{R}\backslash\{-1,1\}$. The range of f on $\{x \in \mathbb{R} \mid -1 < x < 1\}$ is $\{y \in \mathbb{R} \mid y \leq -1\}$ and the range of f on $\{x \in \mathbb{R} \mid x < -1\}$ as well as on $\{x \in \mathbb{R} \mid x > 1\}$ is $\{y \in \mathbb{R} \mid y > 0\}$. So overall the range of f is $\{y \in \mathbb{R} \mid y \leq -1 \vee y > 0\}$.
The domain of definition of g is $\{x \in \mathbb{R} \mid x \geq 1\}$ and the range of g is $\{y \in \mathbb{R} \mid y \geq 0\}$.

(b) We define $h = f \circ mul_3^{-1} \circ g$. Give an explicit formula for $h(x)$.
Solution: $mul_3^{-1} \circ g(x) = \frac{\sqrt{x-1}}{3}$.
$f \circ mul_3^{-1} \circ g(x) = \frac{1}{(\frac{\sqrt{x-1}}{3})^2-1} = \frac{1}{\frac{x-1}{9}-1} = \frac{9}{x-10}$.
(This equality holds for $\{x \in \mathbb{R} \mid x \geq 1 \wedge x \neq 10\}$.)

8. **Induction** (12.5 points)
Prove using mathematical induction that the following statement is true for all integer numbers $n \geq 1$:

$$\sum_{k=1}^{n}(k-1)^3 = \frac{((n-1)n)^2}{4}$$

Solution: In the base case $n = 1$, both the left- and the right-hand side of this equation are 0.

Now consider the inductive case:

$$\sum_{k=1}^{n+1}(k-1)^3 = \sum_{k=1}^{n}(k-1)^3 + n^3 = \frac{((n-1)n)^2}{4} + n^3 \quad \text{(induction hypothesis)}$$

$$= \frac{n^4-2n^3+n^2}{4} + n^3 = \frac{n^4+2n^3+n^2}{4} = \frac{(n(n+1))^2}{4}$$